FactLens

Detecting Fake News & Political Bias with Explainable Machine Learning

CAP 5610 · Group 9

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The Problem

The Challenge

- Misinformation spreads 6× faster than truth on social media
- 75% of people cannot reliably identify fake news
- Political bias is invisible to most readers — same facts, different framing
- Manual fact-checking is slow, expensive, and doesn't scale
- Billions of articles published daily — impossible to review manually

Research Questions

RQ1 Can a simple, interpretable pipeline reliably detect fake news from text?

RQ2 Can a model detect political bias (Left vs Right) from article language?

RQ3 Do fake news and political bias share overlapping linguistic fingerprints?

We need automated, explainable ML - not just accurate, but interpretable.

Summary: Real or Fake & Left or Right?

ARTICLE A

Trump equated mail-in voting with cheating.

FAKE 95.8% confident

Why FactLens caught it:

president +0.10	share +0.09	media +0.07

Emotional framing, conspiracy language, no verifiable sources, excessive punctuation — all signals learned from 38,590 training articles.

The model also flagged: informal tone, rhetorical questions, political targeting- linguistic fingerprints of misinformation.

> "Mail-in voting means mail-in cheating."

ARTICLE B

Trump Roils NATO as Pressure Builds Over Strait of Hormuz

LEFT-LEANING 78% confident

Key signals:

nato	alarmed	unpredictable

Left-leaning framing emphasizes diplomatic friction, alliance concerns, and critical language around Trump's foreign policy decisions.

Words like "alarmed", "unpredictable", and "eroding" signal editorial perspective even in factual reporting.

> Trump roils NATO as pressure builds over Strait of Hormuz
> Ellen Mitchell and The Hill 1 day ago

•In a world of headlines, our model separate signal from noise.

Methodology

D1

Fake vs Real News

Source: Kaggle — Bisaillon

44,898 articles

23,481 Fake / 21,417 Real

After deduplication: 38,590 usable articles

D2

Political Bias

Source: Kaggle — cl0ud0

13,366 articles

7,803 Left / 5,563 Right

Binary classification: Left vs Right

Preprocessing Pipeline:

Lowercase + Punctuation → **Remove Stopwords** → **Lemmatization (NLTK)** → **Remove Duplicates** → **TF-IDF Vectorize** → **80/20 Train/Test Split**

Additional features extracted from raw text: sentiment score · subjectivity · readability · exclamation count · question count

Finding: Fake articles use question marks and exclamation marks 12× more than real articles.

Methodology

Model 1 — Logistic Regression

Input:	TF-IDF matrix (50,000 word features) + 5 linguistic features
Why LR?	Fast, interpretable — coefficients directly explain predictions
Class weights:	balanced — handles 54/46 Real/Fake imbalance
Explainability:	Top coefficient words show exactly what drove each prediction
Debiasing:	Source bias experiment: removed 30 Reuters formatting words

Model 2 — DistilBERT

Input:	Cleaned article text via DistilBERT tokenizer
Architecture:	67M parameters, pre-trained on billions of sentences
Fine-tuning:	3 epochs, A100 GPU, 3 min 6 sec training time
Advantage:	Understands full context, negation, and sentence meaning
Explainability:	Integrated Gradients — word-level attribution scores

LR: interpretable by design · DistilBERT: contextual understanding · Both applied to BOTH tasks

Methodology

Explainability Layer

LR Coefficients

Each word gets a weight. Positive → Fake. Negative → Real. Top 20 words extracted and visualized per class.

LIME (attempted)

Near-zero scores — model too confident (>99.99%). Documented as known limitation of perturbation-based methods.

Integrated Gradients

Gradient-based attribution on DistilBERT. 'reuters' scored 0.86 for REAL — single strongest source bias signal.

Cross-Task Analysis

Coefficient Correlation

Pearson $r = -0.049$ ($p=0.15$, not significant). Spearman $\rho = -0.17$. Two tasks are linguistically orthogonal.

Vocabulary Overlap

Zero words shared in top-20 Fake vs top-20 Left. Zero in top-20 Real vs top-20 Right. Models learned different vocabularies.

Linguistic Feature Comparison

KS tests across Fake/Real/Left/Right: mixed results — no consistent political lean in fake news. Confirms independence.

Fake news ≠ political bias — two independent problems that need two independent models.

Key Results - Conclusion

Model	Fake vs Real	Left vs Right
LR — Original (TF-IDF only)	98.52%	
LR — Debiased (source bias removed)	97.50%	
LR — Enhanced (+5 features)	98.33%	87.1%
DistilBERT — Fine-tuned	99.97%	86.6%

Source bias: 'reuters' was LR's #1 Fake indicator - model learned writing style, not truth

DistilBERT beats LR on Fake/Real (+1.5%). On Left/Right both tie - bias is keyword-driven

Fake news & political bias are orthogonal (Pearson $r \approx 0$) - two independent linguistic phenomena

Live Demo

Paste any article — get instant Fake/Real + Left/Right results

Fake vs Real | Left vs Right | Key Word Explanations

Powered by Logistic Regression + TF-IDF · Trained on 52,000+ articles

FactLens

What we built. What we found. What it means.

- Two independent classifiers — fake news detection (99.97%) and political bias (87.1%)
- Source bias is real: LR learned Reuters formatting, not misinformation — debiasing confirmed
- DistilBERT > LR for fake news; LR ≈ DistilBERT for political bias (bias is keyword-driven)
- Fake news and political bias are orthogonal ($r \approx 0$) — they are two separate linguistic problems

Question?